DATA ANALYSIS STRATEGY & PLANNING

Project FOODFLOW

Decoding production efficiency in virtual food processing with advanced Excel

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Date: 14 October 2026
Primary tool: Microsoft Excel (Power Query, PivotTables, dynamic arrays, what-if tools)

THE PLAN AT A GLANCE

33 hrs estimated workload across 8 phases	**5** public data sources, cross-checked	**6** research questions, 4 testable hypotheses	**5** data-quality gates, borrowed from HACCP

The big idea

A food factory is a production line: raw material comes in, gets inspected, processed, packed, and dispatched. **This project runs the analysis itself like that line**, with five "stations" that turn messy public data into clean, trusted, decision-ready insight. Quality gates borrowed from food safety (HACCP) protect the data the same way they protect the product.

1. Project Introduction

1.1 What is "virtual food processing"?

Food processing turns raw agricultural inputs (grain, milk, oilseeds, fruit, meat) into safe, shelf-ready products. **Virtual food processing** means studying and optimising that chain through data and simulation rather than on a physical shop floor: a "digital twin lite" of a processing plant, built from public statistics and tested in a spreadsheet sandbox before any real money or machinery is touched.

For this project, I model a fictional mid-sized plant, **NutriLine Foods**, whose inputs, throughput, labour, energy use and margins are benchmarked against publicly reported industry data. The plant is illustrative, but the numbers it is compared with are real.

1.2 Why data-driven decisions matter here

- **Thin margins.** Food manufacturing earns modest margins, so small efficiency gains (less waste, less energy per tonne) matter a great deal.
- **Volatile inputs.** Crop yields, weather, and commodity prices swing, and processors must plan procurement and production around that volatility.
- **Perishability and safety.** Spoilage, yield loss and compliance failures are expensive. Data helps spot problems before they become recalls or write-offs.
- **Rising pressure on energy and sustainability.** Energy intensity and waste are now both a cost line and a reporting requirement.

1.3 Why Excel?

Excel is the most widely used analysis tool in operations and finance teams. Modern Excel (Power Query, dynamic arrays, Power Pivot, what-if tools) can clean, model, forecast, and visualise a project of this size end to end, and every step stays transparent and auditable, which is exactly what a planning exercise needs.

2. Objectives, Research Questions and Hypotheses

2.1 Analysis objectives

ID	Objective (SMART)
O1	**Consolidate** at least five public datasets into one cleaned, documented Excel data model covering a minimum of 10 years where data allows.
O2	**Quantify trends** in production volume and processing output across five food categories (cereals, dairy, edible oils, fruit and vegetables, meat), including seasonality and volatility.
O3	**Build a Production Efficiency Index (PEI)**: a single composite score combining labour productivity, energy intensity and yield, so segments and years can be ranked fairly.
O4	**Test four hypotheses** about what drives efficiency, using correlation, regression and lag analysis inside Excel.
O5	**Deliver an interactive dashboard and at least five evidence-backed recommendations** for the NutriLine Foods virtual plant, each with a what-if value estimate.

2.2 Key research questions

ID	Research question
RQ1	How have production volumes of key food categories moved over the last 10 to 15 years, and which categories are the most seasonal or volatile?
RQ2	Which food-manufacturing segments create the most value added per worker and per unit of energy?
RQ3	Do raw-material price changes squeeze processing margins, and if so, after what delay?
RQ4	Does higher capital intensity (investment per worker) translate into higher productivity?
RQ5	Where is the largest efficiency gap between NutriLine Foods and industry benchmarks, and what would closing it be worth?
RQ6	What does the next three years of output look like under base, optimistic, and stressed scenarios?

2.3 Working hypotheses

ID	Hypothesis	How Excel will test it
H1	Capital-intensive segments produce more output per worker.	Scatter plot with trendline, CORREL, and LINEST regression (capital per worker vs output per worker).
H2	Raw-material price spikes reduce processing margins with a lag of one to two periods.	Lagged CORREL on offset ranges; conditional formatting to highlight months where price rose, and margin fell.
H3	Segments with higher energy use per tonne score lower on overall efficiency.	PEI ranking table, Data Analysis ToolPak regression, and a colour-scaled heat map.
H4	Staple categories (cereals, oils) are more stable than perishable ones (fruit and vegetables, dairy).	Coefficient of variation per category using STDEV.S / AVERAGE, compared in a PivotChart.

3. Scope and Limitations

In scope	Out of scope
Publicly available annual and monthly data on food production, prices, and food-manufacturing performance. Three lenses: India, the United States, and the European Union, for comparison. Five food categories, plus an "all food manufacturing" aggregate. A fictional virtual plant (NutriLine Foods) used as the benchmark target. All analysis, modelling and dashboards built in Excel.	Proprietary or plant-level operational data. Real-time sensor (IoT) data and live production feeds. Machine-learning or predictive models outside Excel. Food-safety laboratory results and regulatory compliance audits. Detailed financial modelling of real companies.

Limitations to be honest about

- **Aggregated, not plant-level.** Public data describes industries and countries, so NutriLine Foods is a calibrated illustration and not a prediction of any real plant.

- **Definitions differ across sources.** Industry classifications, units, and currencies vary, so harmonisation (see data-quality gates) is necessary and introduces assumptions.
- **Reporting lags.** Official manufacturing surveys are often published with a delay of a year or more, so the latest period may be missing for some sources.
- **Excel scale limits.** A worksheet holds just over one million rows. Large source files will be filtered at import (Power Query) rather than loaded whole.
- **Correlation is not causation.** Findings will be framed as evidence and hypotheses to investigate, not proof.

4. Data Source Identification

Five public sources were chosen so that each research question has at least two independent sources behind it. Together they cover **what is produced** (volumes), **what it costs** (prices), and **how well factories convert inputs into outputs** (efficiency).

Source	What it provides	Why it was selected	Used for
FAOSTAT (UN FAO) fao.org/faostat	Production quantities, yields, and producer prices by country and commodity, with long annual history.	The global standard for food production data. Free, consistent across countries, and downloadable as CSV.	RQ1, RQ3, RQ6 and H2, H4
US Census Bureau: Annual Survey of Manufacturers census.gov/programs-surveys/asm	Food manufacturing (NAICS 311): value of shipments, value added, employment, payroll, capital expenditure, energy and materials cost.	The richest public view of how factories convert inputs to outputs. It supplies the labour, capital, and energy measures behind the PEI.	RQ2, RQ4, RQ5 and H1, H3
India: Annual Survey of Industries (MoSPI) and data.gov.in data.gov.in	Factory-level aggregates for food product manufacturing: output, workers, fixed capital, fuel and materials.	Brings an Indian lens, relevant to a fast-growing processing sector, and allows a developing-economy versus developed-economy comparison.	RQ2, RQ4, RQ5
Eurostat (Structural Business Statistics and PRODCOM) ec.europa.eu/eurostat	Food, beverage manufacturing turnover, employment, production value and volumes by EU country.	Provides a third regional benchmark with harmonised definitions across many countries, which strengthens cross-checks.	RQ2, RQ5, RQ6
Open Food Facts world.openfoodfacts.org	Crowd-sourced product database: categories, ingredients and nutrition for packaged food worldwide.	Adds a product-mix layer to the story, so segments can be profiled by product type and processing level.	Context and segment profiling (RQ2)

Selection logic

Official first. Government and UN sources lead, because their methods are documented and their data can be cited.

Triangulate. Efficiency measures from the US, India, and EU are compared, so no single country's definitions drive the conclusions.

Free and Excel-friendly. Every source offers CSV or XLSX download, so the full pipeline stays reproducible.

Check before use. Table names, release dates, and download formats change over time, so exact tables and licence terms will be confirmed and logged in the workbook during Phase 2.

5. Methodology: The FOODFLOW Line

The analysis moves through five stations, just like a product moving down a processing line. Each station has a clear output, and nothing moves forward until it passes its gate.

Station	Food-line analogy	What happens in the analysis
1. Intake	Receiving raw material	Download datasets, record source, date and licence, and load everything through Power Query into one workbook.
2. Inspect	Quality control checks	Run the five data-quality gates (below). Reject, repair or flag anything that fails.
3. Process	Cooking and transforming	Standardise units and currency, build the master table, calculate ratios and the Production Efficiency Index.
4. Package	Packing and labelling	Create PivotTables, charts, heat maps and an interactive dashboard with slicers.
5. Dispatch	Shipping to customers	Turn findings into recommendations with what-if values and a short executive summary.

5.1 Data-quality gates (inspired by HACCP)

HACCP identifies "critical control points" where food safety must be checked. The same idea applies to data: a few well-placed checks catch most errors before they reach a chart.

Gate	Question asked	Excel check	If it fails
CCP1	Is the data complete?	=COUNTBLANK(range)/ROWS(range) must be under 5% per key column.	Fill using a documented method, or exclude the series.
CCP2	Is it consistent?	Units converted to tonnes and currencies converted to USD in Power Query. Category names mapped through a lookup table.	Fix the mapping table and refresh.
CCP3	Is it plausible?	Z-score formula flags values beyond three standard deviations. Conditional formatting highlights them in red.	Verify against the source. Keep, correct, or annotate.

Gate	Question asked	Excel check	If it fails
CCP4	Is it traceable?	Every row carries Source and Download_Date columns. A Log sheet records every change.	Block the data from the master table until sourced.
CCP5	Is it repeatable?	All cleaning is done as Power Query steps, never by hand edits.	Rebuild the step in Power Query.

5.2 Step-by-step method

1. **Frame.** Confirm objectives, research questions, and hypotheses. Agree what "efficient" means (the PEI components and weights).
2. **Collect.** Download the five datasets, record their source, date, and licence in a README sheet.
3. **Load and clean.** Import with Power Query. Remove duplicates, unpivot wide year columns, standardise units, currencies and category names.
4. **Inspect.** Run the five data-quality gates and record results in the Log sheet.
5. **Build the master table.** Merge sources into one Excel Table keyed by Year, Region and Segment.
6. **Engineer indicators.** Calculate output per worker, value added per worker, energy per tonne, material cost share, yield, growth rates and volatility.
7. **Compute the PEI.** Normalise each component to a 0 to 1 scale, apply weights held in named assumption cells, and rank segments and years.
8. **Explore and test.** Build PivotTables for trends and seasonality, then test H1 to H4 with CORREL, LINEST and the Data Analysis ToolPak.
9. **Forecast and stress-test.** Use FORECAST.ETS for three-year outlooks, and Data Tables and Scenario Manager for base, optimistic, and stressed cases.
10. **Optimise (what-if).** Use Goal Seek and Solver to estimate the value of closing NutriLine's efficiency gaps.
11. **Visualise.** Assemble the dashboard: KPI cards, slicers, trend lines, heat map and gap-closure waterfall.
12. **Conclude.** Write findings, recommendations and limitations. Peer-check formulas and numbers.

6. The Advanced Excel Toolkit

Tool or technique	Where it is used	Why it earns its place
Power Query	Import, unpivot, merge, and clean all five sources.	Repeatable steps, so refreshing with new data takes one click.
Excel Tables and structured references	Master data table and every calculation sheet.	Formulas read like sentences and expand automatically with new rows.
XLOOKUP / INDEX-MATCH	Category mapping, benchmark lookups.	Two-way, error-tolerant lookups with no column-order fragility.
SUMIFS / AVERAGEIFS / COUNTIFS	Ratios by region, segment and year.	Multi-criteria aggregation without pivot dependence.
Dynamic arrays (FILTER, SORT, UNIQUE, LET)	Live leaderboards and top-N segment lists.	Results spill and update automatically, with no copy-down formulas.

Tool or technique	Where it is used	Why it earns its place
PivotTables, PivotCharts, Slicers and Timelines	Trends, seasonality and segment comparison.	Fast exploration, and the dashboard becomes interactive.
Conditional formatting	Outlier flags, heat maps, icon-set KPIs, data bars.	Patterns and problems become visible at a glance.
Data validation	Assumption cells and dashboard selectors.	Prevents invalid inputs from breaking the model.
Statistical functions and ToolPak	CORREL, STDEV.S, LINEST, regression output.	Tests the hypotheses with real statistics.
FORECAST.ETS and trendlines	Three-year outlooks with seasonality.	Time-series forecasting without leaving Excel.
Data Tables, Scenario Manager, Goal Seek, Solver	Base, optimistic and stressed scenarios, and gap-closure value.	Turns analysis into decisions: "what if we cut energy use by 10%?"
Sparklines and charts	Compact trend lines in KPI cards and scatter plots with trendlines.	Dense, readable communication of results.

6.1 Signature formulas

Examples of the formulas that sit at the heart of the model (cell references are illustrative):

```
Normalised score (0 to 1):   =(C2-MIN(C$2:C$40))/(MAX(C$2:C$40)-MIN(C$2:C$40))
PEI (weights = named cells):  =w_labour*N2 + w_energy*O2 + w_yield*P2
Volatility (CV):            =STDEV.S(D2:D16)/AVERAGE(D2:D16)
Lagged correlation (1 per): =CORREL(D3:D40, E2:E39)
Outlier flag (CCP3):        =ABS((C2-AVERAGE(C$2:C$40))/STDEV.S(C$2:C$40))>3
Live top-5 leaderboard:     =TAKE(SORT(FILTER(Master,Master[Year]=sel_yr),7,-1),5)
Three-year forecast:        =FORECAST.ETS(A41,B2:B40,A2:A40,1)
```

The PEI weights (for example, 40% labour productivity, 30% energy efficiency, 30% yield) are deliberately held in labelled input cells, so the sensitivity of the ranking to those choices can itself be tested.

6.2 Dashboard design

- **Top row:** four KPI cards (average PEI, output per worker, energy per tonne, forecast growth) with icon-set arrows and sparklines.
- **Controls:** slicers for Year, Region and Segment, plus a timeline for monthly data.
- **Centre:** trend line for production volume, colour-scaled heat map of PEI by segment and year.
- **Bottom:** scatter of capital per worker against output per worker (H1) and a waterfall chart showing the value of closing each NutriLine gap.

7. Timeline and Resource Planning

The plan totals **33 hours**, inside the 30 to 35 hour estimate. It is built for **10 working sessions of about three hours each**, which can run over two to three weeks. Two hours are held back as a buffer in Phase 8, and the larger phases (P3, P5) have slack inside them.

7.1 Task breakdown

Phase	Key tasks	Hours	Deliverable
P1. Frame and plan	Confirm objectives, questions, hypotheses, PEI definition. Set up workbook structure and README.	2	Approved scope and workbook skeleton
P2. Intake: find and collect data	Locate exact tables, download five datasets, record source, date and licence, and run a first look at each file.	5	Raw data sheets and source log
P3. Inspect and clean	Power Query cleaning, unit and currency standardisation, category mapping, and CCP1 to CCP5 checks.	5	Clean master table and quality log
P4. Process: model and indicators	Engineer ratios, normalise, build the PEI, add NutriLine benchmark profile.	4	Calculation sheets and PEI ranking
P5. Analyse and test	PivotTables, seasonality and volatility, correlation and regression for H1 to H4, forecasts and scenarios.	7	Findings sheet and hypothesis verdicts
P6. Package: dashboard	Build charts, heat maps, slicers, KPI cards, and a what-if panel. Test the interactions.	4	Interactive dashboard
P7. Dispatch: insights and report	Write insights, recommendations with what-if values, limitations, and executive summary.	4	Final report and recommendations
P8. Quality assurance and buffer	Formula audit, number cross-check, peer feedback, polish and contingency.	2	Verified, submission-ready workbook

7.2 Timeline view (D = working session)

Phase	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Hours
Frame and plan											**2**
Intake: find and collect data											**5**
Inspect and clean											**5**
Process: model and indicators											**4**
Analyse and test											**7**
Package: dashboard											**4**
Dispatch: insights and report											**4**
Quality assurance and buffer											**2**
Total											**33**

7.3 Resources

Resource	Detail
Software	Microsoft Excel (365 recommended for dynamic arrays, Power Query, and Power Pivot). Enable the Data Analysis ToolPak and Solver add-ins.
Data	Five free public sources listed in Section 4. All are downloadable as CSV or XLSX.
People	One analyst (me), plus a peer or mentor for a review in Phase 8.
Documentation	README and Log sheets inside the workbook, with this plan as the governing document.
Cost	No data or licensing cost.

8. Risk Assessment

Each risk is scored for likelihood (L) and impact (I) on a 1 to 3 scale. Priority = L x I, so anything scoring 6 or more gets an active mitigation from day one.

Risk	L	I	Priority	Mitigation
Inconsistent definitions across sources	3	3	**9**	Build the category mapping table early, document every assumption, and compare at index level (growth and ratios) as well as at absolute level.
Missing or delayed data	3	2	**6**	Set a 5% completeness threshold (CCP1), use documented interpolation only for short gaps, and exclude series that fail rather than guess.
Spreadsheet formula errors	2	3	**6**	Use Tables and named ranges, separate inputs from calculations, add check cells, and have a peer audit in Phase 8.
Large files slow Excel	2	2	4	Filter and aggregate in Power Query before loading, and use the Data Model rather than raw sheets for big tables.
Currency and inflation distortions	2	3	**6**	Convert to a single currency, use real (inflation-adjusted) values for trend analysis, and note the base year.
Over-reading correlations	2	3	**6**	State findings as evidence, not proof. Check the lags, test with more than one source, and show the r-squared.
Scope creep	2	2	4	Hold to the five questions and five sources. Park extra ideas in a "Phase 2 ideas" list.
Time overrun	2	2	4	Use the 2-hour buffer, timebox each phase, and cut dashboard extras before cutting quality checks.
Data licence or access changes	1	2	2	Keep a dated copy of each download in the workbook and note the licence terms in the README.

9. Expected Outcomes and Success Criteria

9.1 What the finished project will deliver

- A single, documented Excel workbook that anyone can refresh with new data.

- A Production Efficiency Index that ranks segments and years transparently.
- A clear verdict (supported, partly supported, not supported) on each of the four hypotheses.
- An interactive dashboard and five or more recommendations, each with a what-if value.

9.2 How success will be judged

Criterion	Test
Comprehensive	Every research question is answered with data from at least two sources, and every objective has a visible output.
Relevant	Every tab and technique ties back to food processing efficiency and a named question.
Clear	A reader can follow the logic from question to method to result without a guide, and the dashboard can be understood in under two minutes.
Original	HACCP-style data gates, a custom efficiency index, a fictional virtual plant benchmarked against real data, and what-if valuation of every recommendation.
Reproducible	Refreshing Power Query with new data updates every table and chart without manual repair.

> **Closing thought**
>
> Good analysis planning is a recipe, not a guess: know what you are cooking (objectives), what goes into it (data), how you will prepare it (method), how long it takes (timeline), and what could burn (risks). This plan puts all five on the table before the first formula is typed.

Appendix: Workbook Blueprint

The planned structure of the Excel file, so the build stays organised from day one.

Sheet	Purpose
README	Project summary, source list with dates and licences, sheet guide, and version notes.
Assumptions	PEI weights, base year, currency rates, and scenario inputs, all as named cells with data validation.
Raw_FAO / Raw_ASM / Raw_ASI / Raw_EU / Raw_OFF	Untouched downloads, one sheet per source. Never edited by hand.
Map	Lookup tables that map categories, countries and units across sources.
Master	Clean, merged Excel Table keyed by Year, Region, and Segment.
Calc_PEI	Indicators, normalisation and the Production Efficiency Index.
Tests	Correlation, regression, lag analysis, and hypothesis verdicts.
Scenarios	Forecasts, Data Tables, Goal Seek and Solver what-if models.
Pivots	PivotTables and PivotCharts feeding the dashboard.
Dashboard	KPI cards, slicers, charts and heat map, presented for the reader.
Log	Data-quality gate results and a record of every data decision.